Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/23/2007
PI Name:	Lee, Te-Won Ph.D.		
Project Title:	Automatic Speech Recognition in Noisy Environments - Human Factors Approach		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tlee@softmax.com	**Fax:**	FY 619-839-3119
PI Organization Type:	INDUSTRY	**Phone:**	858-736-5300
Organization Name:	SoftMax, Inc		
PI Address 1:	6885 Flanders Drive, Suite A		
PI Address 2:			
PI Web Page:	www.softmax.com		
City:	San Diego	**State:**	CA
Zip Code:	92121	**Congressional District:**	49
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	03/10/2003	**End Date:**	04/10/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Lee		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1479		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of the third year was the optimization, evaluation and testing of the ICA technology developed by SoftMax, Inc. for speech enhancement in noisy shuttle (ISS) environment. To achieve the task goals, we have chosen several commercial recognition systems where its recognition rate has been further enhanced by our ICA technology. We have tested the proposed cofiguration with human subjects at JSC on the Robonaut task. In ISS voice recognition capability enhances the human performance. Accurate and reliable recognition is necessary to deploy the system. Our technology yields significant improvements over the existing technology.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The focus of the application has been on Robonaut Command and Control via voice recognition. Test results have shown significant increase in voice recognition accuracy using the ICA technology.
Task Progress:	The speech recognition system has been further optimized with actual subject testing data.
Bibliography Type:	Description: (Last Updated: 06/13/2005)